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Coupling EDFM-AI and ESMDA for Dynamic Unconventional Reservoir Characterizations

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Abstract

Hydraulic fracturing is the key enabling technology for the economic development of unconventional reservoirs, where well productivity depends on the creation of an extensive, conductive fracture network. Accurate characterization of effective fracture geometry and reservoir properties is essential for predicting well performance and optimizing stimulation design. However, conventional historymatching approaches in these complex systems are hindered by high-dimensional uncertainty, strong nonlinearity, and the high computational cost of large-scale simulations. This study presents an integrated workflow that couples a GPU-accelerated Embedded Discrete Fracture Model (EDFM) with the Ensemble Smoother with Multiple Data Assimilation (ESMDA) and AI-based surrogate modeling to perform automatic, uncertainty-aware history matching in unconventional reservoirs. The EDFM provides high-fidelity representation of complex hydraulic and natural fractures, GPU acceleration enables rapid evaluation of large ensembles, and AI surrogates reduce the number of full-physics simulations required per assimilation cycle. The workflow systematically assimilates production and pressure data to estimate both static and time-dependent parameters, while quantifying uncertainty through posterior parameter distributions. The approach is validated using two benchmark cases: (1) a single well with a single hydraulic fracture and time-dependent fracture permeability decline, and (2) a multi-layer, three-well configuration with multiple uncertain parameters, including fracture and matrix properties. In both cases, the workflow achieved accurate history matches and recovered the true parameter values with relative errors typically below 15%, while requiring only seconds of wall-clock time per simulation on a GPU. Results demonstrate that the proposed EDFM-ESMDA-AI framework enables scalable, automated, and physically consistent dynamic reservoir characterization, offering a practical solution for field-scale unconventional reservoir modeling.

Introduction

Hydraulic fracturing has become the cornerstone technology for unlocking hydrocarbons from unconventional reservoirs, enabling the economic production of oil and gas from ultra-low-permeability formations such as shale, tight sandstone, and coal seams. In these plays, well productivity and ultimate recovery are largely dependent on the creation of an extensive, conductive, and well-connected fracture network that maximizes contact between the wellbore and the reservoir matrix. As such, accurate characterization of effective fracture geometry, including half-length, height, conductivity, and connectivity, is critical for predicting well performance, optimizing stimulation design, and improving field development strategies.

Although a range of field diagnostic techniques such as microseismic monitoring, pressure transient testing, and fiber-optic distributed sensing can provide valuable but incomplete insights into fracture behavior, production history matching remains one of the most practical and widely adopted approaches for quantifying effective fracture properties in unconventional reservoirs. By calibrating a numerical reservoir model against observed production and pressure data, engineers can infer fracture and reservoir characteristics that are otherwise difficult or impossible to measure directly. However, history matching in unconventional reservoirs is inherently challenging. The coexistence of heterogeneous rock properties, complex hydraulic-natural fracture interactions, and multiphase flow processes leads to a large, nonlinear, and high-dimensional parameter space. Traditionally, this process has relied on manual parameter tuning, an iterative and time-consuming procedure in which engineers adjust uncertain reservoir and fracture parameters until simulated results align with historical data. This approach is computationally expensive and prone to non-uniqueness, especially in multi-well settings with long production histories.

Recent advances in numerical modeling and data assimilation have provided new opportunities to address these limitations. The Embedded Discrete Fracture Model (EDFM) offers a high-fidelity approach for modeling fracture-flow interactions by explicitly representing fractures as discrete planar features embedded within a structured reservoir grid (Moinfar et al., 2013; Panfili and Cominelli, 2014; Shakiba and Sepehrnoori, 2015; Cavalcante Filho et al., 2015; Jiang and Younis, 2016; Du et al., 2017; Xu et al., 2017, Xu et al., 2018, 2019; Chen et al., 2020; Sepehrnoori et al., 2020). This approach avoids the excessive mesh refinement required in conventional local grid refinement methods while maintaining numerical accuracy in handling complex geometries, including intersecting, non-planar, and multi-scale fractures. EDFM is therefore particularly well suited to unconventional reservoirs, where fracture complexity strongly governs fluid flow. At the same time, ensemble-based data assimilation algorithms such as the Ensemble Smoother with Multiple Data Assimilation (ESMDA) have proven effective in systematically calibrating uncertain model parameters by assimilating field measurements over multiple assimilation steps. By updating an ensemble of realizations, ESMDA reduces mismatch between simulated and observed data while preserving parameter variability, enabling robust uncertainty quantification alongside parameter estimation.

Despite these advancements, directly coupling EDFM with ESMDA for unconventional reservoir characterization remains computationally demanding, particularly when applied to large ensembles, multi-well models, and long production periods. This computational barrier limits the practical deployment of such workflows in field-scale applications. The integration of artificial intelligence and high-performance computing offers a transformative solution to this challenge. GPU-based parallelization can drastically accelerate EDFM simulations, allowing multiple realizations to be evaluated simultaneously, while AI-driven surrogate modeling can further reduce computational cost by approximating reservoir responses for unexplored parameter combinations, thus decreasing the number of full-physics simulations required in each assimilation cycle.

The present study develops an AI-integrated EDFM-ESMDA workflow for automatic and dynamic characterization of unconventional reservoirs. The workflow combines the physical fidelity of EDFM-based full-physics simulation with the statistical rigor of ESMDA, enhanced by GPU reservoir simulation acceleration and AI-assisted surrogate modeling. The goal is to automate the history matching process, minimize human intervention, accelerate simulations, and quantify uncertainties in both reservoir and fracture properties. The approach is validated using benchmark cases with varying geological complexity, demonstrating its capability to provide fast, robust, and uncertainty-aware dynamic characterization. The novelty of this work lies in the seamless integration of EDFM-GPU simulation, ESMDA-based parameter assimilation, and AI-enhanced computational acceleration into a single framework, enabling scalable application to large-scale, multi-stage, multi-well unconventional reservoir development scenarios.

Methodology

The proposed workflow integrates an Embedded Discrete Fracture Model (EDFM)-AI hybrid simulation framework with the Ensemble Smoother with Multiple Data Assimilation (ESMDA) algorithm to perform automatic history matching for unconventional reservoirs, as shown in Figure 1. The coupling leverages EDFM's capability to efficiently represent complex fracture geometries and GPU acceleration for high-speed simulation, while ESMDA systematically assimilates production and pressure data to update reservoir and fracture parameters. This process is iterative, with each cycle refining the uncertain parameters to reduce mismatch between simulated and observed field data. The AI component accelerates the workflow by providing fast surrogate predictions and intelligent parameter search guidance, thereby reducing the number of full physics runs needed. The detailed steps of this proposed workflow is provided below:

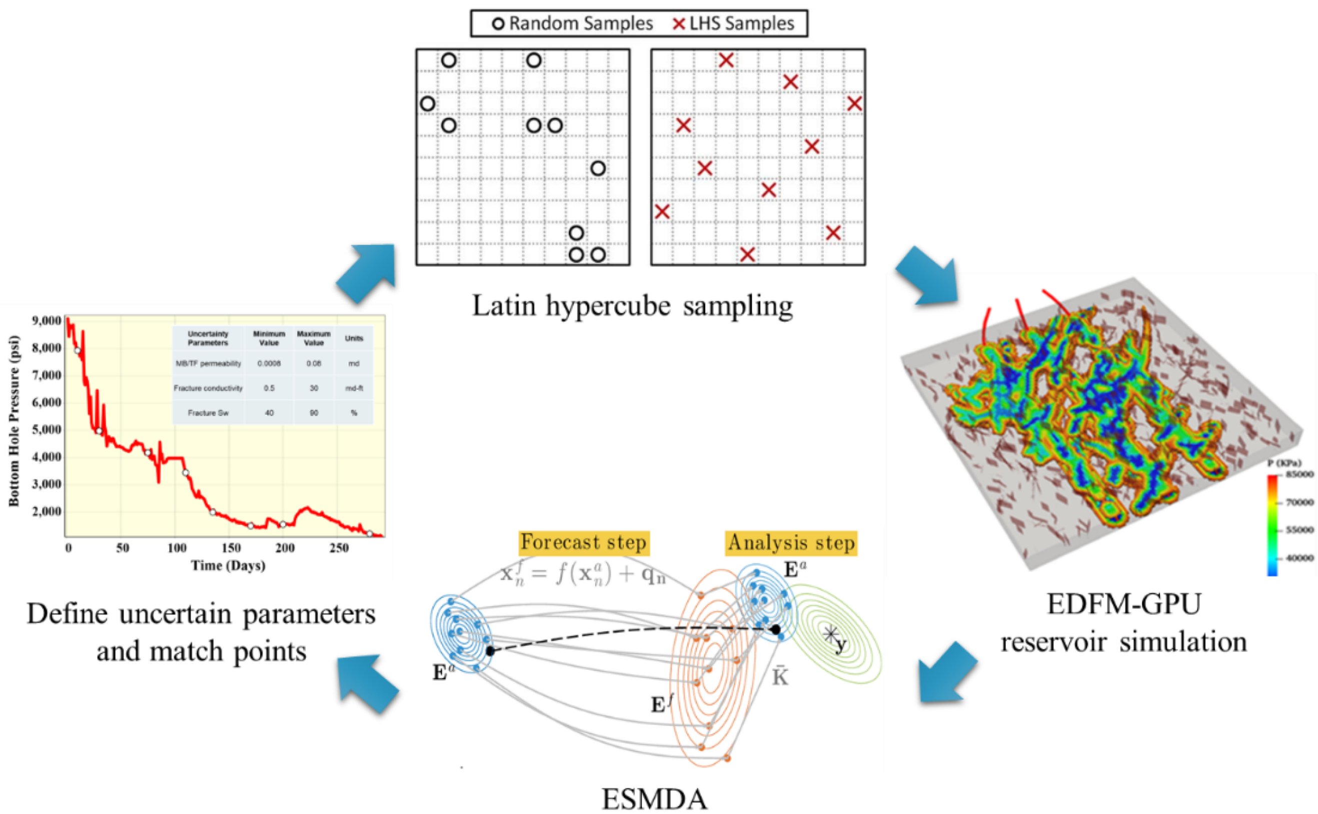


Figure 1—EDFM-AI-ESMDA workflow for automatic history matching for unconventional reservoirs.

Step 1: Define uncertainty parameters and matching points. The first step involves identifying and parameterizing the sources of uncertainty that significantly impact production response. These parameters can include reservoir properties: matrix permeability, porosity, saturation etc. Fracture properties: fracture half-length, height, conductivity, and natural fracture density etc. Simultaneously, a set of match points is defined. These are typically discrete timesteps corresponding to available historical measurements such as daily or monthly oil/gas/water rates and bottomhole pressure. The match points serve as the locations where simulated results will be compared with historical data during each assimilation cycle.

Step 2: Generate initial parameter samples. To explore the parameter space efficiently, Latin Hypercube Sampling (LHS) is used to generate the initial ensemble of model realizations. Given **Np** uncertain parameters and **Ne** ensemble members, LHS stratifies each parameter's range into **Ne** intervals and randomly selects one sample from each interval without replacement, ensuring good coverage of the entire multi-dimensional uncertainty space.

Step 3: Run EDFM-GPU reservoir simulations. For each ensemble member, the EDFM and GPU reservoir simulator are executed to generate the predicted responses at the predefined match points. The EDFM component efficiently represents both hydraulic and natural fractures without the need for fine grid refinement, while GPU-accelerated reservoir simulation enables rapid solution of the multi-physics coupled subsurface fluid flow system.

Step 4: Perturb observations. To account for measurement uncertainty, the historical data vector is perturbed by adding noise consistent with the observation error covariance matrix. This stochastic perturbation prevents ensemble collapse and allows for robust uncertainty quantification. The equation for observation perturbation is shown in Eqn.1:

$$\mathbf{d}_{obs,j}^{(k)} = \mathbf{d}_{obs} + \epsilon_j^{(k)},\ \epsilon_j^{(k)} \sim N\left(0,\ \alpha_k \mathbf{C}_d\right) \qquad \text{Eqn. 1}$$

where $\mathbf{d}_{obs,j}^{(k)}$ is the perturbed observations, d_{obs} is the original observations, $\epsilon_j^{(k)}$ is the perturbation error drawn from a multivariate normal distribution N, C_d is the data error covariance matrix representing uncertainties in the observed data, a_k is an inflation factor applied at assimilation step k in order to spread the assimilation over multiple iterations.

Step 5: Compute cross-covariance and auto-covariance. The ESMDA algorithm requires estimation of the cross-covariance (Eqn. 2) between model parameters and simulated data, and the auto-covariance of simulated data (Eqn. 3). These matrices quantify how variations in parameters influence model outputs and how outputs are correlated with each other.

$$C_{\theta G} = \frac{1}{N-1}\sum_{i=1}^{N}\left(\theta_i - \overline{\theta}\right)\left(d_i^{pred} - \overline{d}^{pred}\right)^T \qquad \text{Eqn. 2}$$

$$C_{GG} = \frac{1}{N-1}\sum_{i=1}^{N}\left(d_i^{pred} - \overline{d}^{pred}\right)\left(d_i^{pred} - \overline{d}^{pred}\right)^T \qquad \text{Eqn. 3}$$

where $C_{\theta G}$ is the cross-covariance, C_{GG} is the auto-covaraince, N is the number of ensemble members (how many realizations), θ_i is the model parameter vector of ensemble member (vector of uncertainty parameters), $\overline{\theta}$ is the ensemble mean of the parameters, d_i^{pred} is the predicted data from the forward model for ensemble member i (for example simulated bottomhole pressure at match points), $\overline{d}^{pred}$ is the ensemble mean of the predicted data. T is the reverse matrix operation.

Step 6: Parameter update (ESMDA assimilation). The ensemble members are updated using the ESMDA formula. This update process adjusts parameters in the direction that reduces mismatch while respecting prior uncertainty. The update function is shown in Eqn. 4:

$$\theta_i^{(k+1)} = \theta_i^{(k)} + K^{(k)}\left(d_{obs} + \sqrt{\alpha_k}\,\epsilon_i - G\left(\theta_i^{(k)}\right)\right) \qquad \text{Eqn. 4}$$

where $\theta_i^{(k+1)}$ is the updated parameter vector of ensemble member i after the assimilation, $\theta_i^{(k)}$ is the parameter vector of ensemble member i before update, $K_{(k)}$ is the Kalman gain matrix which is equal to $C_{\theta G}\left(C_{GG} + \alpha_k C_d\right)^{-1}$, ϵ_i, is a random vector representing observational noise for ensemble member *i, i*, $G\left(\theta_i^{(k)}\right)$ is the predicted data from the forward model using current parameters $\theta_i^{(k)}$

Step 7: Iteration until convergence. Steps 3-6 are repeated until a maximum number of assimilation cycles is reached, or the objective function (e.g., normalized least-squares mismatch) falls below a user-defined tolerance. The final updated ensemble represents the posterior distribution of parameters, providing both calibrated parameter estimates with the associated uncertainties.

Field Case Studies and Benchmark Validation

To evaluate the performance, accuracy, and robustness of the proposed EDFM-ESMDA workflow, two benchmark case studies were designed and executed. The first case represents a single-well scenario with one hydraulic fracture, which provides a controlled environment for self-validation against a known truth model. The second case extends the complexity to a multi-well, multi-layer configuration, allowing assessment of the workflow's capability in handling multiple interacting wells with variable geological properties. Both cases were simulated using the GPU-accelerated EDFM reservoir simulator, with ESMDA employed for parameter estimation. The objective in each case was to match bottomhole pressure (BHP) histories under fixed production constraints while inferring key uncertain fracture and reservoir parameters.

Case 1: Single Well with a Single Hydraulic Fracture

The first benchmark case involves a single horizontal well intersecting one planar hydraulic fracture, as shown in Figure 2. The key fracture and reservoir properties are summarized in Table 1. The model has a lateral extent of 2,400 ft in the x-direction, 60 ft in the y- direction, and 120 ft in the z-direction, discretized into 160 × 10 × 40 grid blocks. The fracture half-length is set to 1,000 ft, with a fracture height of 100 ft and a conductivity of 4 mD -ft. The relative permeability functions differ between the fracture and matrix domains, as shown in Figure 3, to better reflect the dual-porosity nature of the system.

Table 1—Key reservoir and fracture parameters used in the single-well case

Model parameter	Value	Unit
Number of grid blocks (x × y × z)	160 × 10 × 40	-
Grid size (x × y × z)	15 × 6 × 3	ft
Model size (x × y × z)	2400 × 60 × 120	ft
Fracture half-length	1000	ft
Fracture height	100	ft
Fracture conductivity	4	mD-ft

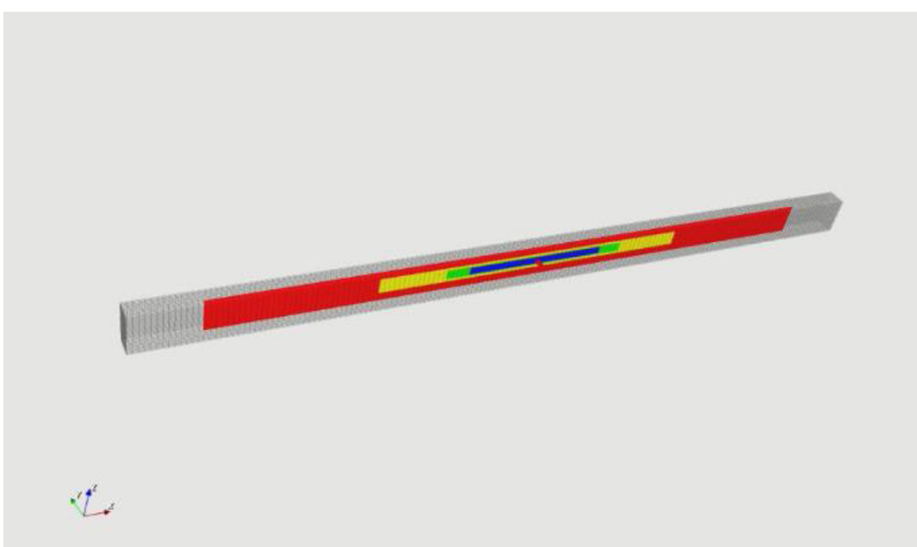


Figure 2—Schematic of the single-well benchmark model showing the well trajectory and fracture placement.

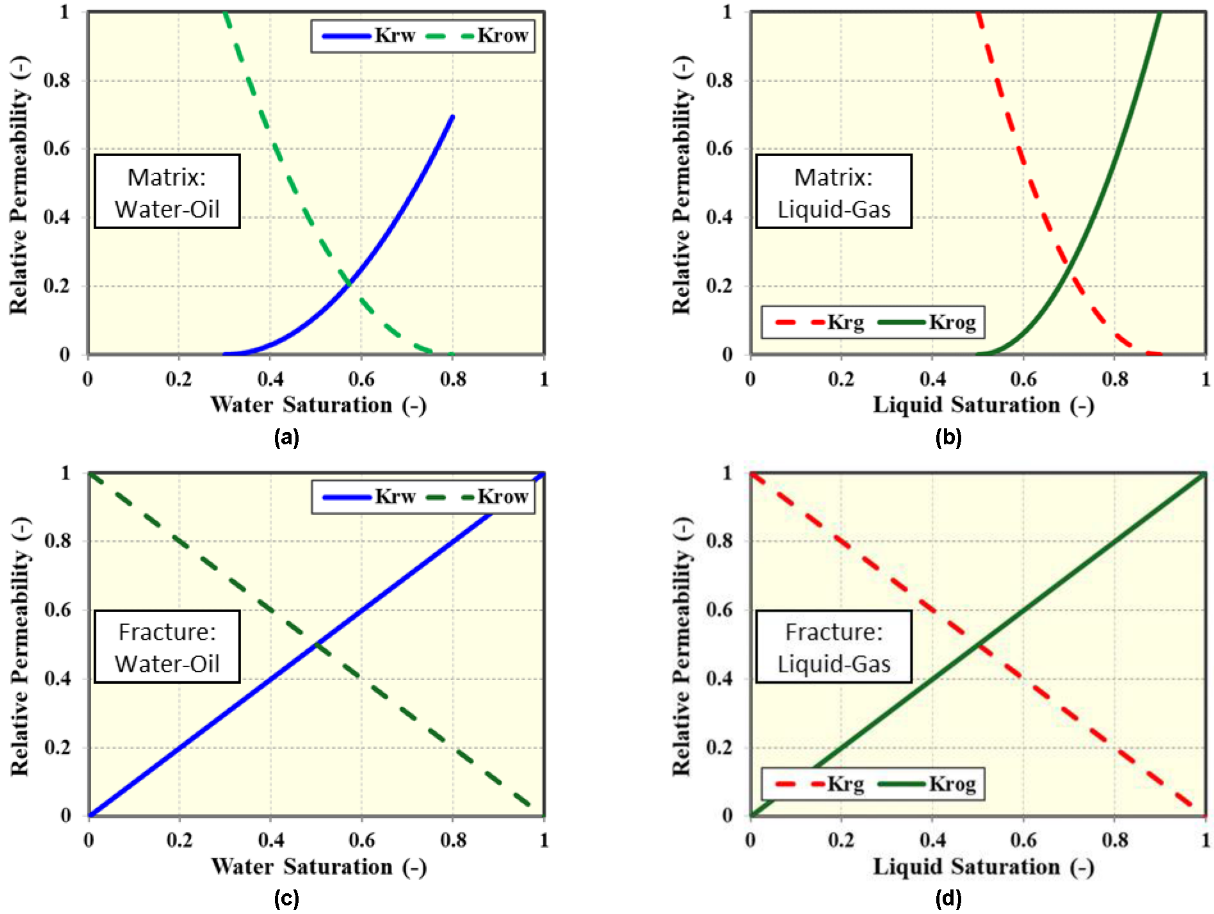


Figure 3—Relative permeability curves for the model: (a) water-oil in the matrix; (b) gas-oil in the matrix; (c) water-oil in the fracture; (d) gas-oil in the fracture.

This case represents a tight oil reservoir, where production is constrained by a fixed oil flow rate. The history matching target is the bottomhole pressure, while the primary uncertainty is the temporal change in fracture permeability. Specifically, the true model assumes permeability declines of 50%, 30%, and 22% at production times of 1, 3, and 6 months, respectively, reflecting proppant crushing. All other parameters remain constant during the simulation.

The GPU-accelerated EDFM simulator achieves a CPU-equivalent runtime of only 26 seconds per simulation. A total of 200 ensemble realizations were generated and simulated. Out of these, 53 models met the convergence criteria for acceptable history matches. Figure 4 shows the comparison between observed and simulated BHP, demonstrating an excellent agreement over the entire production period. Figure 5 further illustrates the inversion of fracture permeability decline behavior, with the EDFM-ESMDA workflow recovering the true permeability change profile with negligible deviation. This result confirms that the proposed approach can accurately capture dynamic fracture conductivity evolution, even with limited data and strong nonlinearity in the response.

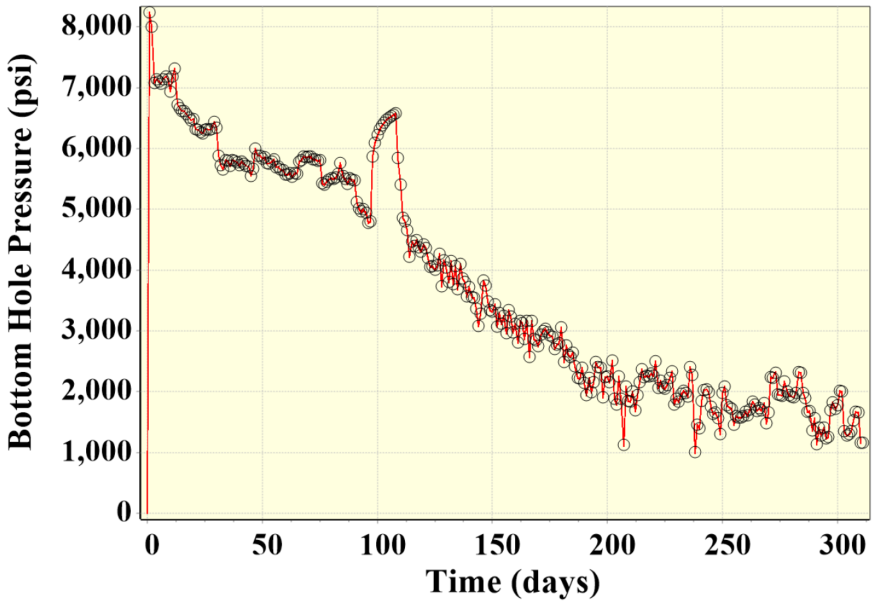


Figure 4—History match results for bottomhole pressure in the single-well case.

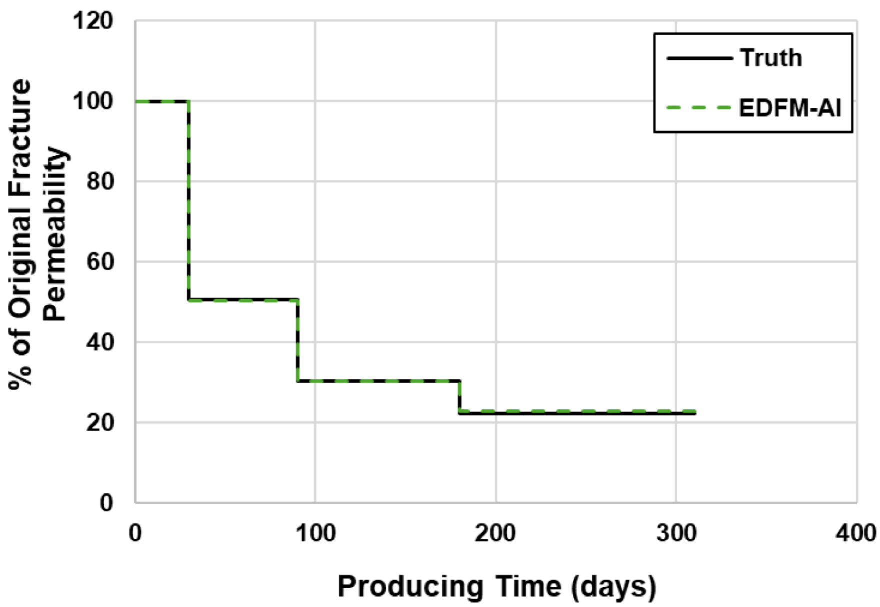


Figure 5—Comparison of true and estimated fracture permeability decline over time.

Case 2: Three Wells with Single Fractures in a Multi-Layer Reservoir

The second benchmark case introduces additional complexity, involving three horizontal wells, each intersecting a single hydraulic fracture (Figure 6). Wells 1 and 3 are landed in the upper reservoir layer, while Well 2 is landed in the lower layer. The two layers have distinct rock properties. Key reservoir and fracture parameters are listed in Table 2. The model spans 3,200 ft * *60* ft * 210 ft, discretized into 160 * 10 * 70 grid blocks. The fracture half-length is 1,000 ft and the height is 100 ft for all wells, but fracture conductivity, initial water saturation, and permeability vary between wells.

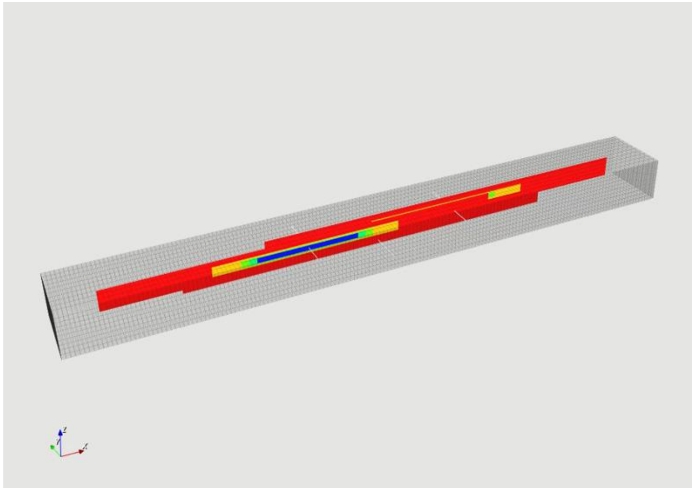


Figure 6—Schematic of the three-well benchmark model with fracture locations.

Table 2—Key reservoir and fracture parameters for the three-well case

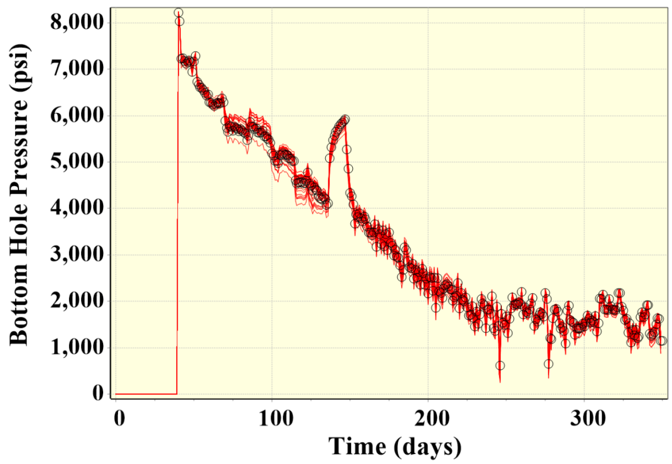
Model parameter	Value	Unit
Number of grid blocks (x × y × z)	160 × 10 × 70	-
Grid size (x × y × z)	20 × 6 × 3	ft
Model size (x × y × z)	3200 × 60 × 210	ft
Fracture half-length	1000	ft
Fracture height	100	ft

For this case, four parameters are treated as uncertain: fracture conductivity, fracture water saturation, matrix permeability, and the time-dependent change in fracture permeability at 1, 2, 3, and 6 months. The true values for each well are given in Table 3. Production is again constrained by fixed oil rates, and BHP is the matching target.

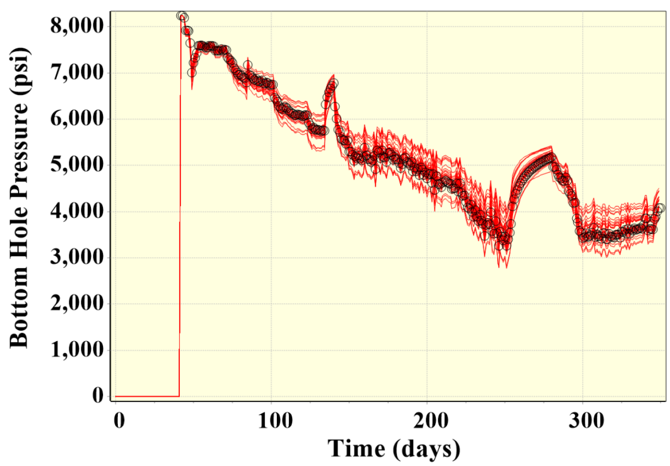
Table 3—True values of uncertain reservoir and fracture parameters for all three wells

Uncertain parameters	Well-1	Well-2	Well-3	Unit
Landing layer	Upper	Lower	Upper	-
Fracture conductivity	4	7	5.5	md-ft
Fracture water saturation	75	65	82	%
Matrix permeability	0.004	0.002	0.004	mD
Fracture permeability change at 1 month	57	47	62	%
Fracture permeability change at 2 months	36	40	45	%
Fracture permeability change at 3 months	23	16	27	%
Fracture permeability change at 6 months	17	19	21	%

The GPU-based reservoir simulator completes each run in 45 seconds of CPU-equivalent time. A total of 100 ensemble realizations were simulated, and 31 realizations were identified as valid history matches. Figure 7 compares the observed and simulated BHP for all three wells, showing excellent agreement across the entire production history. The EDFM-ESMDA workflow successfully recovered the majority of uncertain parameters with relative errors below 15%, as shown in Table 4. The ability to accurately resolve time-varying fracture permeability is particularly noteworthy. Figure 8 compares the true and inverted permeability decline profiles for each well, showing close alignment even in the presence of inter-well interference and heterogeneous rock properties. This confirms that the integrated EDFM-ESMDA approach is not only capable of matching production histories but also of providing physically meaningful parameter estimates.



(a)



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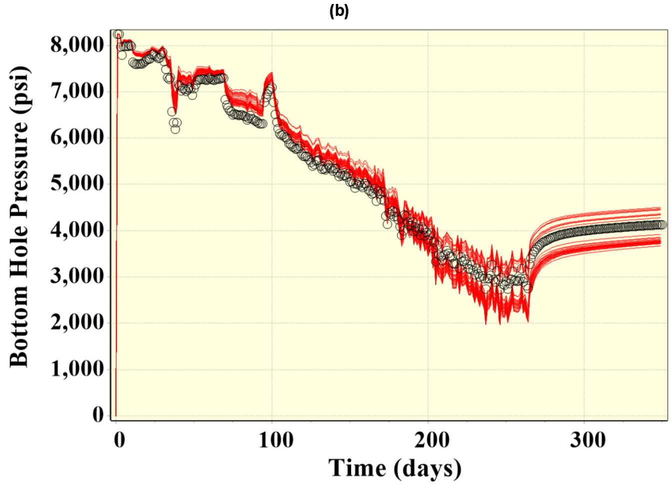

(c)

Figure 7—History match results for bottomhole pressure: (a) Well 1; (b) Well 2; (c) Well 3.

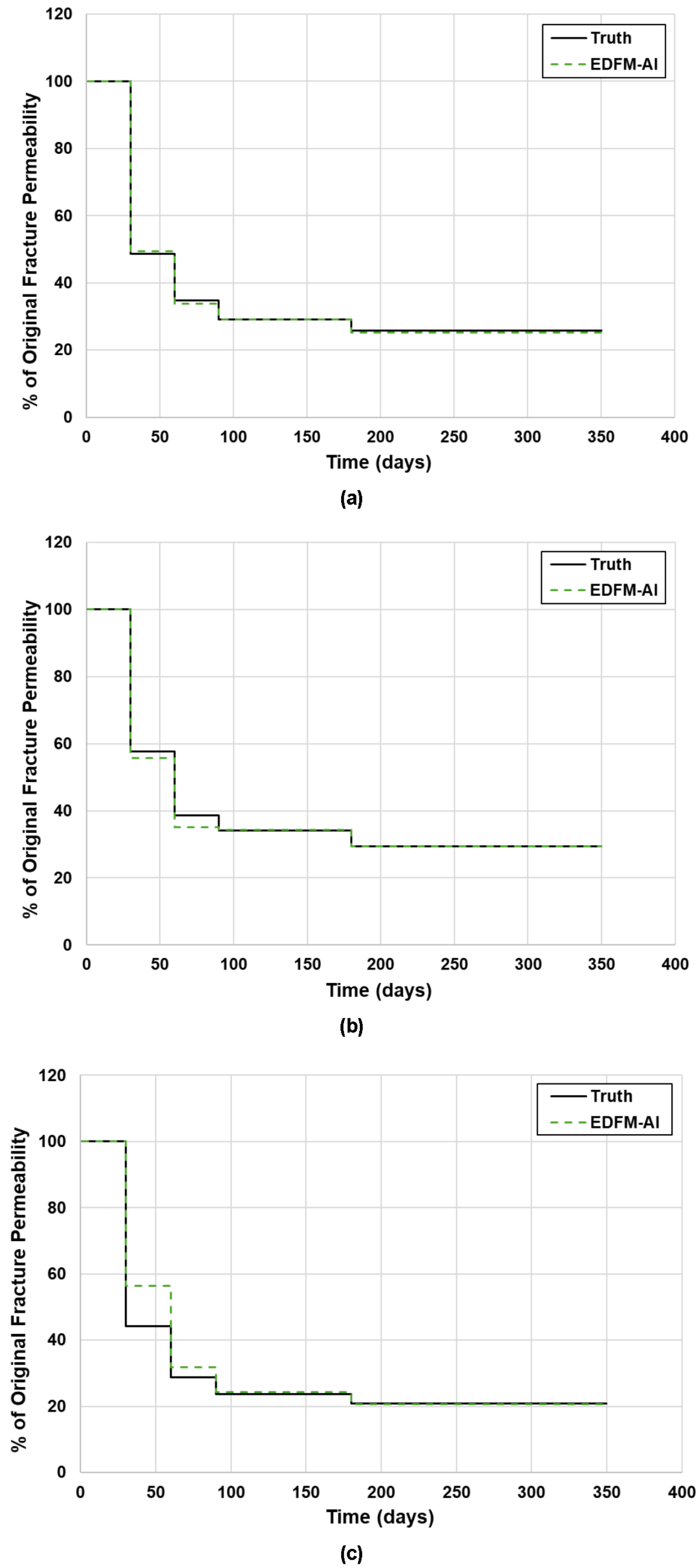

Figure 8—Comparison of true and estimated fracture permeability decline over time for: (a) Well 1; (b) Well 2; (c) Well 3.

Table 4—Comparison between true parameter values and EDFM-ESMDA inversion results for the three-well case, including relative errors

Uncertain parameters	Well-1	Well-2	Well-3	Unit
Landing layer	Upper	Lower	Upper	-
Fracture conductivity (true)	4	7	5.5	md-ft
Fracture conductivity (EDFM-AI)	3.9	6.6	4.9	md-ft
Relative error	1.4	5.5	11.2	%
Fracture water saturation (true)	75	65	82	%
Fracture water saturation (EDFM-AI)	77.8	66.4	73.7	%
Relative error	3.7	2.2	10.7	%
Matrix permeability (true)	0.004	0.002	0.004	mD
Matrix permeability (EDFM-AI)	0.0046	0.0022	0.0046	mD
Relative error	15.1	11.6	15.1	%

Conclusions

This work introduced and validated an AI-integrated EDFM-ESMDA workflow for fast, automatic history matching and uncertainty quantification in unconventional reservoirs. The proposed framework combines the physical accuracy of EDFM for representing complex hydraulic and natural fracture systems, the statistical robustness of ESMDA for ensemble-based parameter assimilation, and the computational efficiency of GPU acceleration and AI surrogate modeling.

Benchmark case studies demonstrate that the workflow can reliably match production histories and accurately estimate both static and dynamic parameters, including time-varying fracture permeability, even in multi-well, multi-layer configurations with heterogeneous rock properties. The single-well case showed that the inversion results matched the true fracture permeability decline profile almost exactly, confirming the method's ability to capture dynamic fracture behavior. The three-well case further validated the robustness of the workflow under inter-well interference and increased parameter dimensionality, with relative errors typically below 15%.

Beyond accuracy, the proposed approach significantly reduces computational cost, achieving CPU-equivalent simulation times of less than one minute per realization and enabling efficient evaluation of large ensembles. The integration of AI surrogate modeling further enhances scalability, making it feasible to apply this workflow to field-scale, multi-stage unconventional reservoir developments.

Future work will focus on extending the method to incorporate additional data sources such as fiber-optic strain sensing, microseismic monitoring, and tracer tests for richer fracture characterization. Further development will also target the integration of geomechanical effects, proppant transport modeling, and real-time data assimilation capabilities to support closed-loop reservoir management. Ultimately, the EDFM-ESMDA-AI workflow has the potential to transform unconventional reservoir characterization by enabling fast, automated, and uncertainty-aware decision-making at both the well and field scale.

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